Cassava mealybugs and green lacewings

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1 Overview

The cassava mealybug arrived in Thailand in 2008 and spread through its cassava fields. Green lacewing larvae eat mealybugs and can be bred and released. Promrak and Rattanakul (2015) built a cellular automaton of a small cassava field. Each plant carries the life stages of mealybugs and lacewings, mealybugs spread to nearby plants and are blown in by wind, infested plants are surveyed and removed, and lacewing larvae are released. They asked how often releases should be made. Promrak and Rattanakul (2017) drew the same model's algorithm as flowcharts, which this implementation follows. It is an adaptation. The sources leave out a few rules, and the release sizes they state do not explain the totals they report.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Statistics
```

2 The model

A 40 × 40 field has one cassava plant per square metre. Each plant is susceptible, infested or removed, and carries real-valued numbers of mealybug eggs, instars and adults and of lacewing eggs, larvae, pupae and adults. Each day:

- within every plant, the life stages follow the source's difference equations, and a lacewing larva eats up to 20 mealybugs of each stage. That is the model's rule.
- In a sweep over the plants in random order, wind brings instars from outside (more often on the two border rows). Infested plants pass instars to plants up to three cells away (with probability 0.05, 0.005 or 0.0005 by ring), and a plant whose mealybugs are all gone is cleared. Every 14 days from day 31 about 60 plants are surveyed, and infested ones found are removed.
- While the last survey found mealybugs, 1000 larvae are released (800 if fewer than half the surveyed plants were infested), split over the infested plants, at the chosen interval.

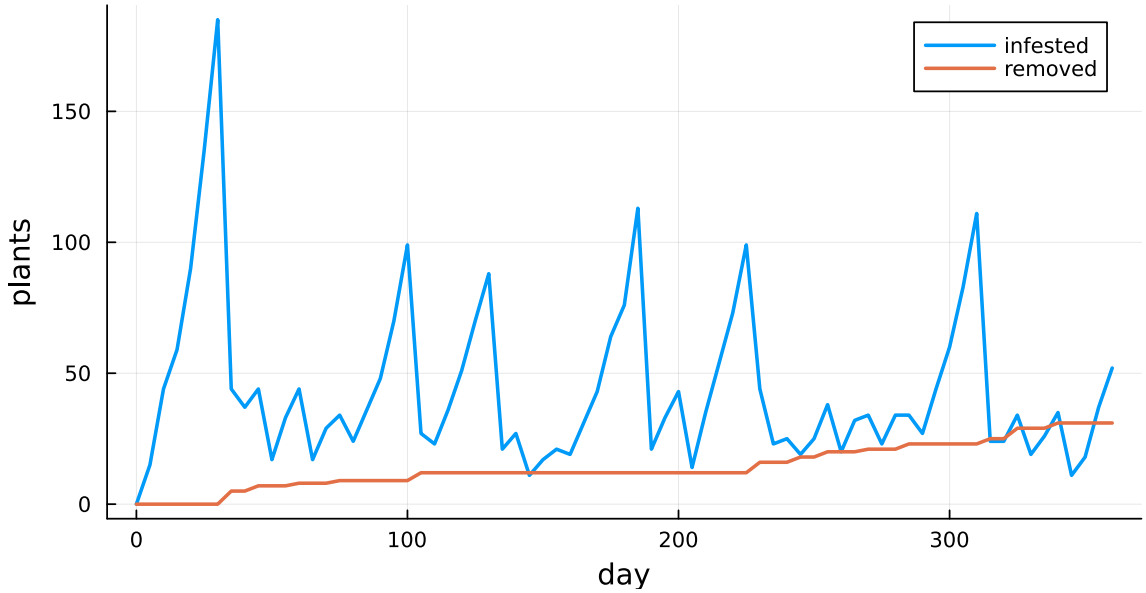
The sweep and the releases need the whole field, so they are a callback stored with the problem. It also keeps the field's memory of releases.

```
prob = problem(:cassava_mealybug; interval=14, seed=1)
keys(prob.kwargs)
```

```
(:callback,)
```

3 A season with releases every two weeks

```
sol = solve(prob; saveat=5)
counts(s) = [count(==(s), u.state) for u in sol.u]
plot(sol.t, [counts(1) counts(2)]; label=["infested" "removed"], lw=2, xlabel="day",
     ylabel="plants", size=(650, 340))
```

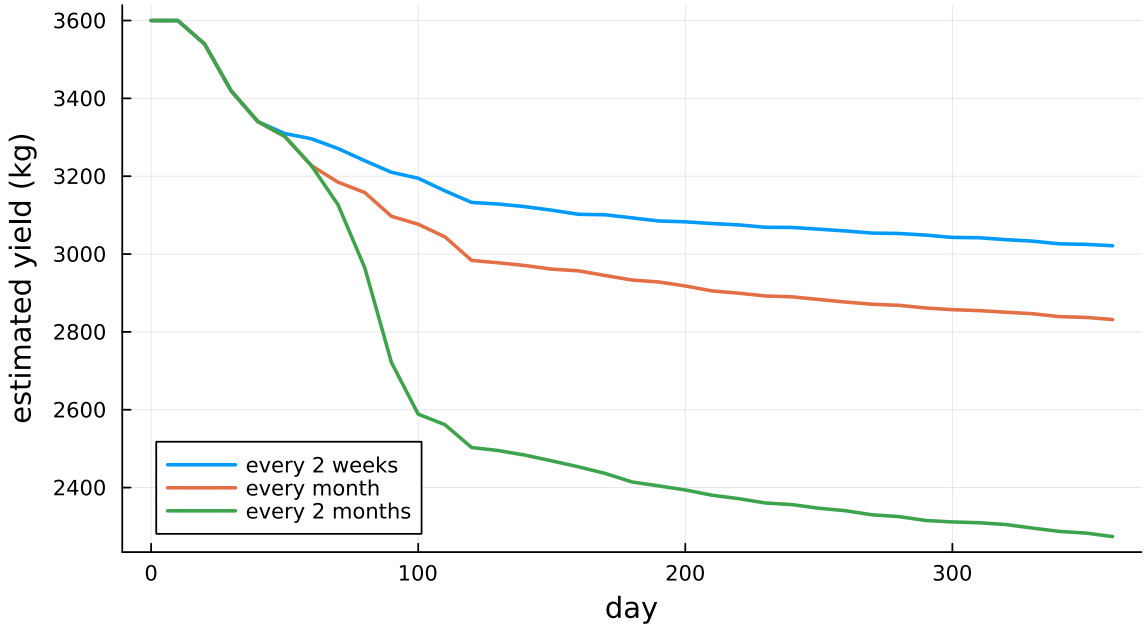


Mealybugs blow in from the first day and spread. Infestation peaks before the first survey, at day 31, which finds them and starts the releases. The larvae clear the infested plants within days. New mealybugs keep arriving, so infestation then rises and falls between about 15 and 110 plants with each round of releases, while removals accumulate at each survey. The source's Figure 3 shows the same course: about 235 plants infested at day 30, a sharp fall at the first release, and 20 to 90 afterwards.

4 Release intervals

The source compared releases every two weeks, every month and every two months. Ten runs each, with the estimated yield over the season. A plant infested for two weeks loses all its yield if this happens in the first four months, three tenths in the next three, and a tenth afterwards:

```
function season(interval; runs=10)
    yields = Vector{Vector{Float64}}()
    released = Float64[]
    for s in 1:runs
        integ = init(problem(:cassava_mealybug; interval, seed=s); saveat=10)
        sol = solve!(integ)
        push!(yields, cassava_yield.(sol.u))
        push!(released, integ.p.released)
    end
    return mean(yields), released
end
intervals = (14, 30, 60)
results = Dict(iv ⇒ season(iv) for iv in intervals)
plot(0:10:360, [results[iv][1] for iv in intervals]; lw=2,
     label=["every 2 weeks" "every month" "every 2 months"], xlabel="day",
     ylabel="estimated yield (kg)", size=(650, 360), legend=:bottomleft)
```




```
[(interval=iv, yield=round(results[iv][1][end]),
released=round(mean(results[iv][2])))
 for iv in intervals]
```

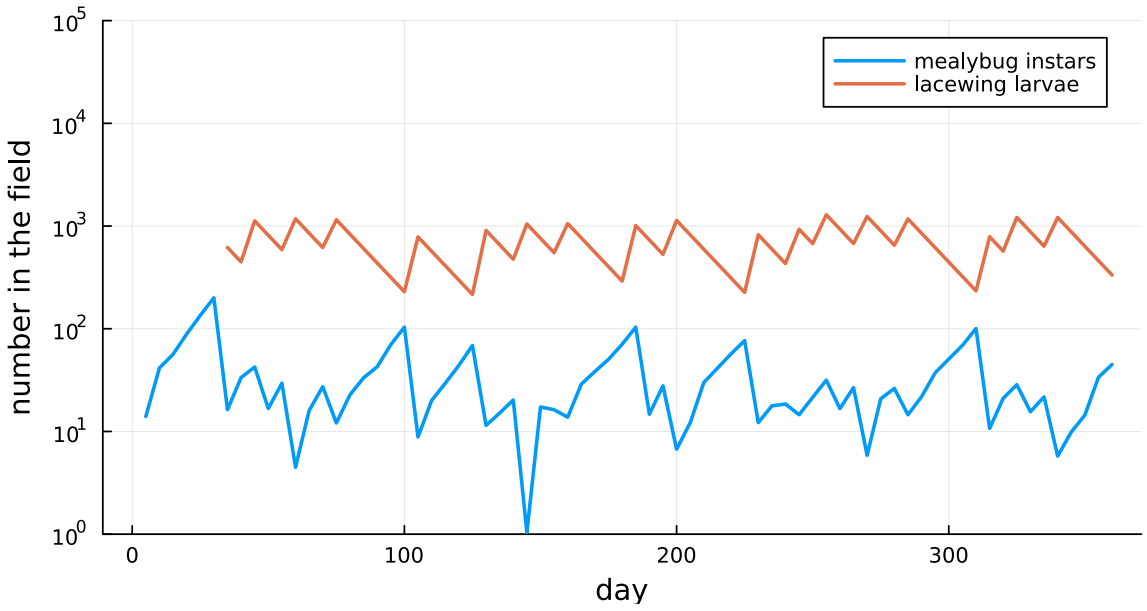
```
3-element Vector{@NamedTuple{interval::Int64, yield::Float64, released::Float64}}:
 (interval = 14, yield = 3021.0, released = 15120.0)
 (interval = 30, yield = 2832.0, released = 9280.0)
 (interval = 60, yield = 2274.0, released = 6000.0)
```

Starting from 3600 kg, the source's yields end near 2960 kg with releases every two weeks and near 2915 kg with the other two intervals, about 50 kg apart. Releases every two weeks agree with the source here. Longer intervals cost much more yield than in the source, because infestation builds up between releases. The source's cumulative releases also climb to about 36 000 larvae in steps of about 3000, the same for all three intervals. That is several times more than here, and more than its stated release sizes of 1000 and 800 explain. The source must have released more, or released at every survey that found mealybugs whatever the interval, but its text does not say.

5 Larvae and mealybugs

The populations over the field, with releases every two weeks. The source's Figure 4 shows instars peaking near 220 before the first release and then between about 10 and 80, and larvae between about 1000 and 3000:

```
plot(sol.t, [[sum(max.(u.Pi, 0)) for u in sol.u] [sum(u.Mi) for u in sol.u]]; lw=2,
     label=["mealybug instars" "lacewing larvae"], xlabel="day", ylabel="number in
the field",
     yscale=:log10, ylims=(1, 1e5), size=(650, 340))
```



The instars follow the source closely. The larvae number a few hundred to about 1200, fewer than in the source, as the releases here are smaller.

6 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:cassava_mealybug).deviations)
```

```
- The model follows the 2015 experiment (40 x 40 field, 30 C, larval releases at
three intervals) with the algorithm drawn in the 2017 paper's flowcharts.
- The mealybug count m1 above which plants surveyed in days 121 to 210 are removed
is not given; by default (m1 = Inf) plants are not removed in those days.
- The survey sample is not given exactly; 60 plants are sampled on each survey day,
about what the 2017 paper's design (every second row, every eleventh plant) gives on
a 40 x 40 field.
- Released larvae are split evenly over the infested plants; the source says only
that they are released at random on infested plants. The source's cumulative
releases rise in steps of about 3000, which its stated release sizes (1000 and 800)
do not explain.
- The last survey's findings are kept until the next survey, and releases follow
every interval days from the first release while they show infestation.
```

```
- Counts are real numbers and, as in the source's equations, are not clamped at
zero, so predation can take a stage below zero; a plant is cleared when its total is
not positive (the source's 'P not > 0'), and its counts are then set to zero. An
infested plant passes on an instar only if it has at least one.
- Predation is min(20 Mi, P) for each mealybug stage, the source's min(20, P/Mi)
times Mi without division by zero.
- The yield class of a plant is the worst it has reached: never infested for two
weeks, or infested for two weeks in days 1-120, 121-210 or 211-360.
```

Promrak, Jirapond, and Chontita Rattanakul. 2015. "Simulation Study of the Spread of Mealybugs in a Cassava Field: Effect of Release Frequency of a Biological Control Agent." *Kasetsart Journal (Natural Science)* 49 (6): 963–70.

Promrak, Jirapond, and Chontita Rattanakul. 2017. "Effect of Increased Global Temperatures on Biological Control of Green Lacewings on the Spread of Mealybugs in a Cassava Field: A Simulation Study." *Advances in Difference Equations* 2017: 161. https://doi.org/10.1186/s13662-017-1218-y.